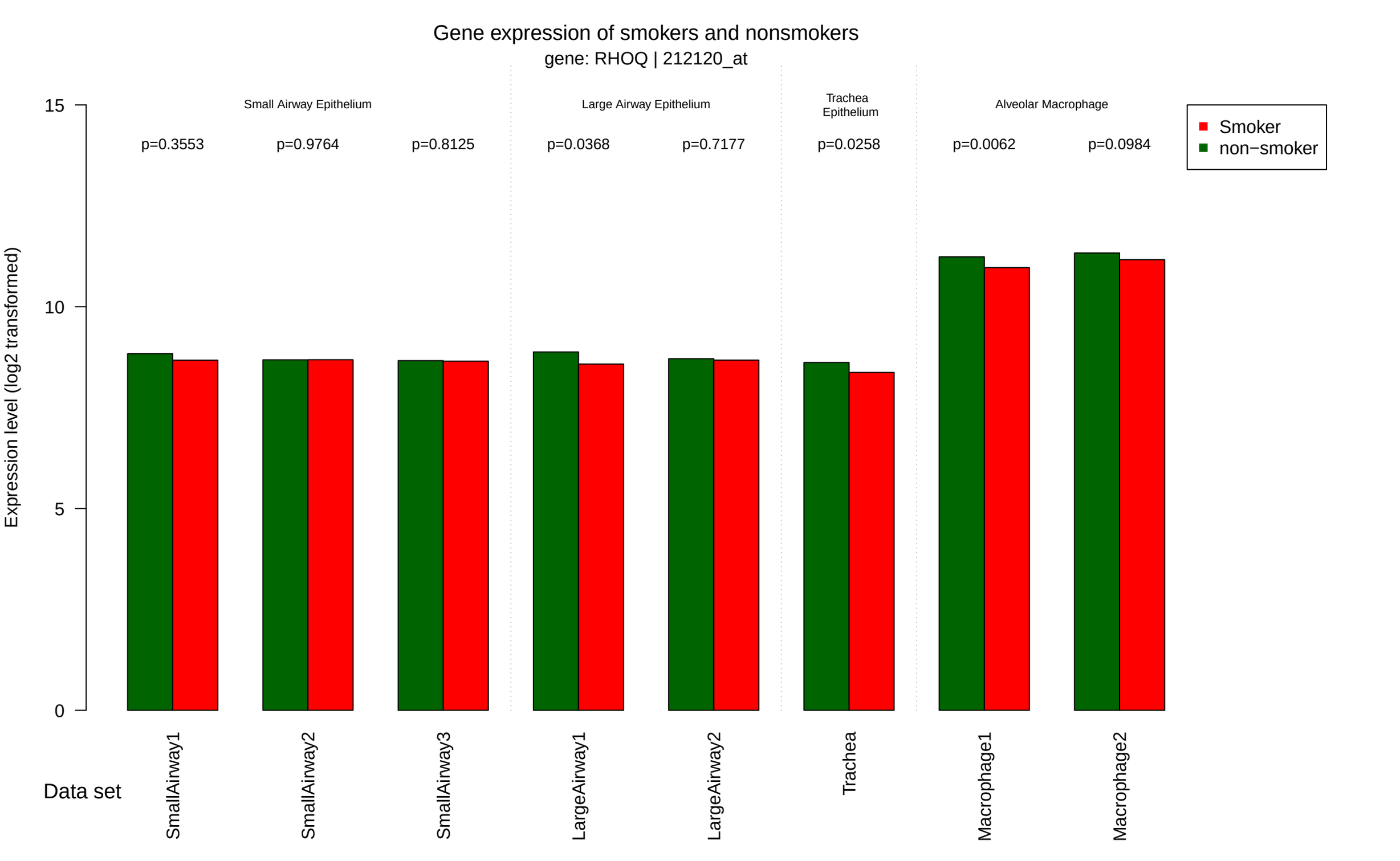
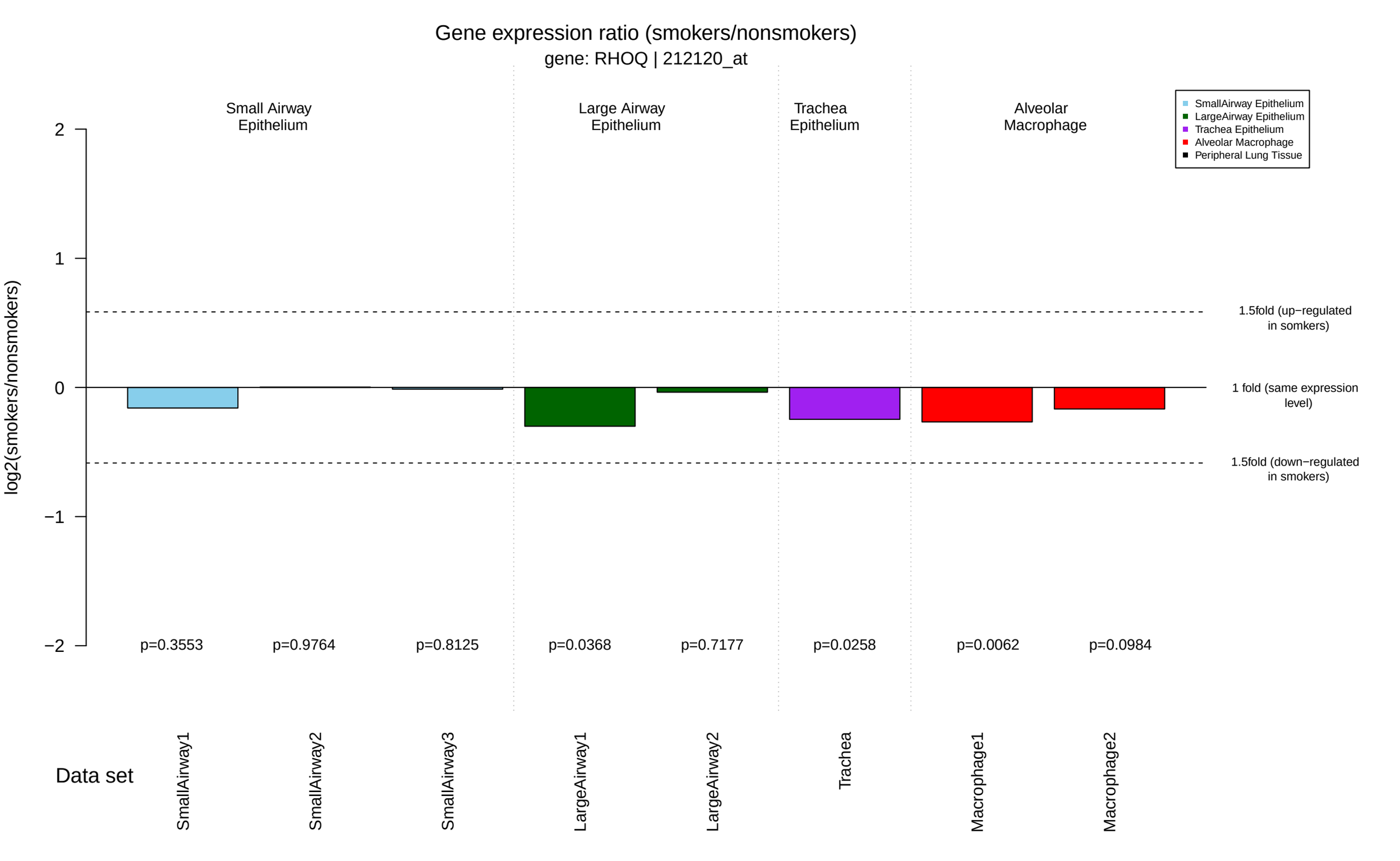
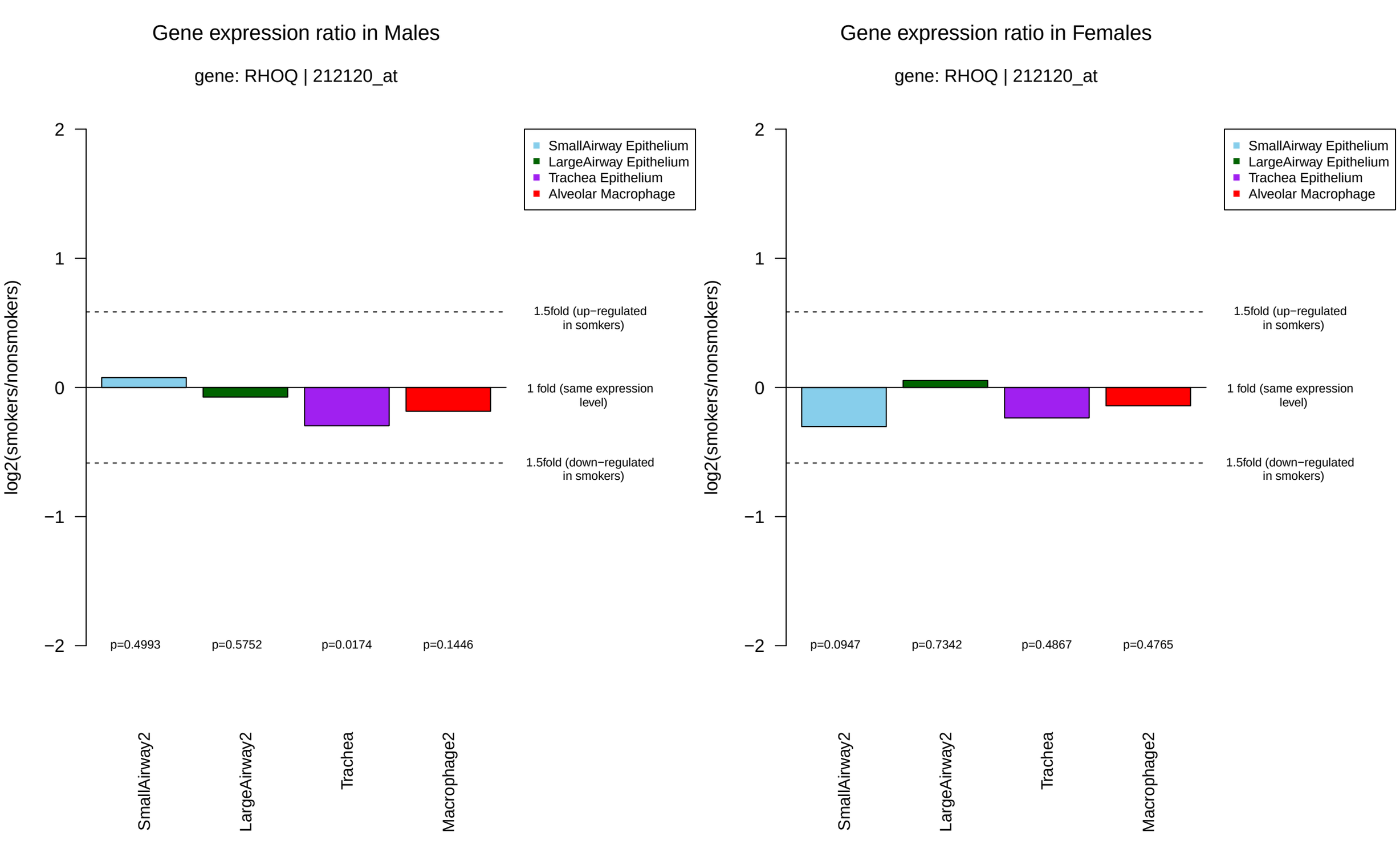
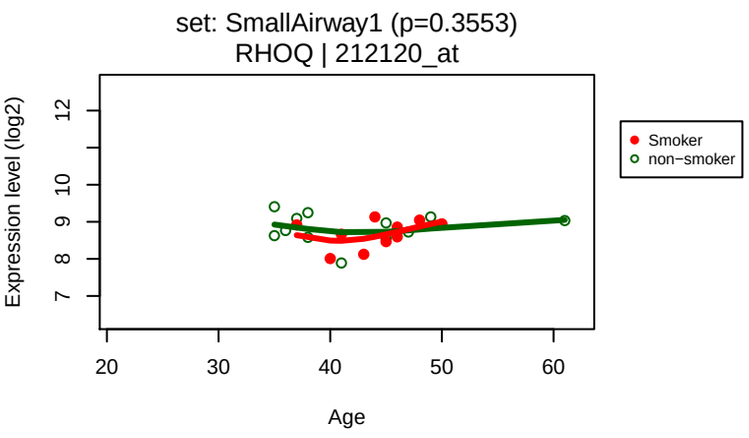
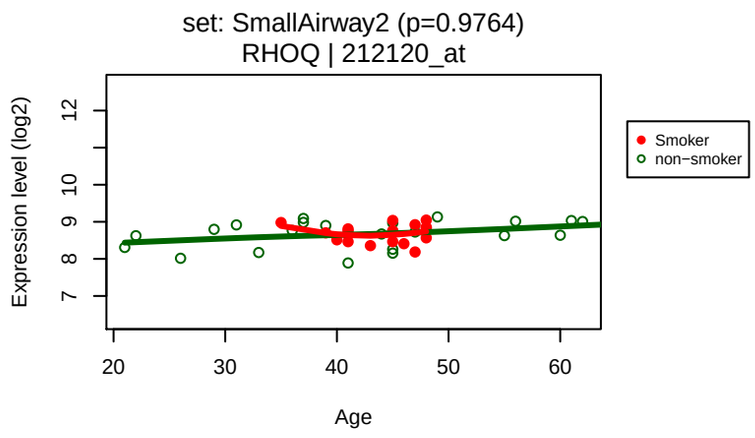
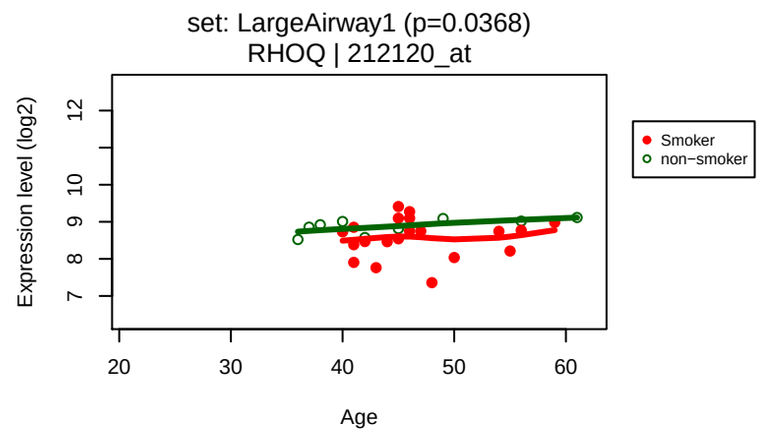

Gene expression of smokers and nonsmokers
gene: RHOQ | 212120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3553
p=0.9764
p=0.8125
p=0.0368
p=0.7177
p=0.0258
p=0.0062
p=0.0984
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

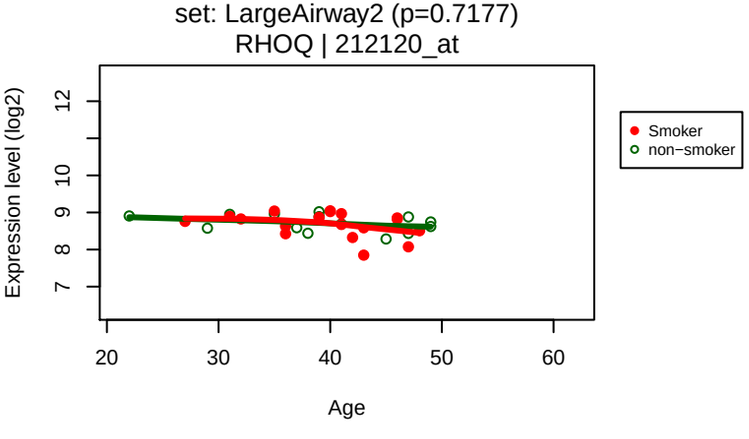

Gene expression ratio (smokers/nonsmokers)
gene: RHOQ | 212120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3553
p=0.9764
p=0.8125
p=0.0368
p=0.7177
p=0.0258
p=0.0062
p=0.0984
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

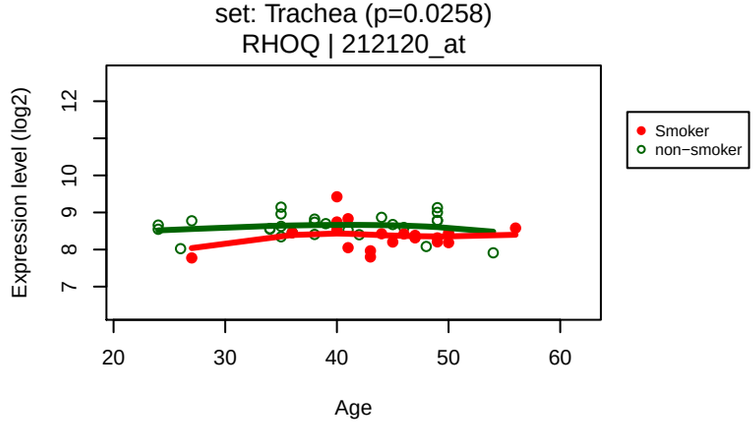

Gene expression ratio in Males
gene: RHOQ | 212120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4993
p=0.5752
p=0.0174
p=0.1446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOQ | 212120_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0947
p=0.7342
p=0.4867
p=0.4765
SmallAirway2
LargeAirway2
Trachea
Macrophage2

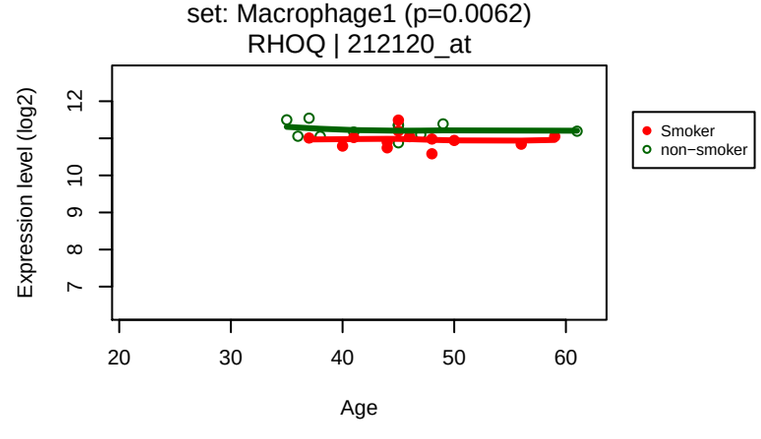

set: SmallAirway1 (p=0.3553)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker

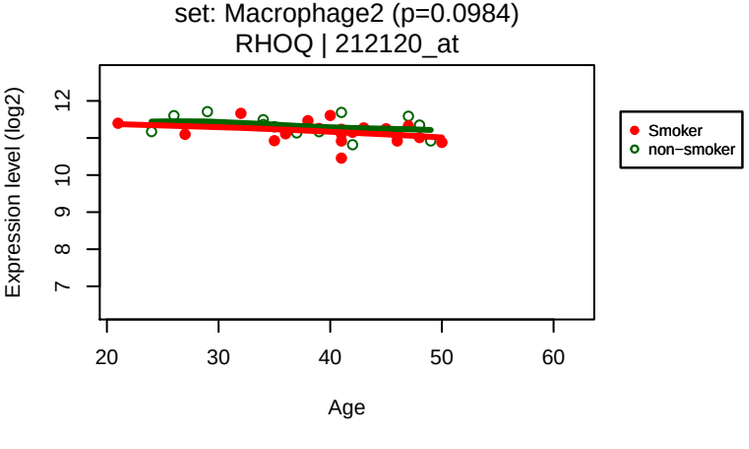

set: SmallAirway2 (p=0.9764)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker

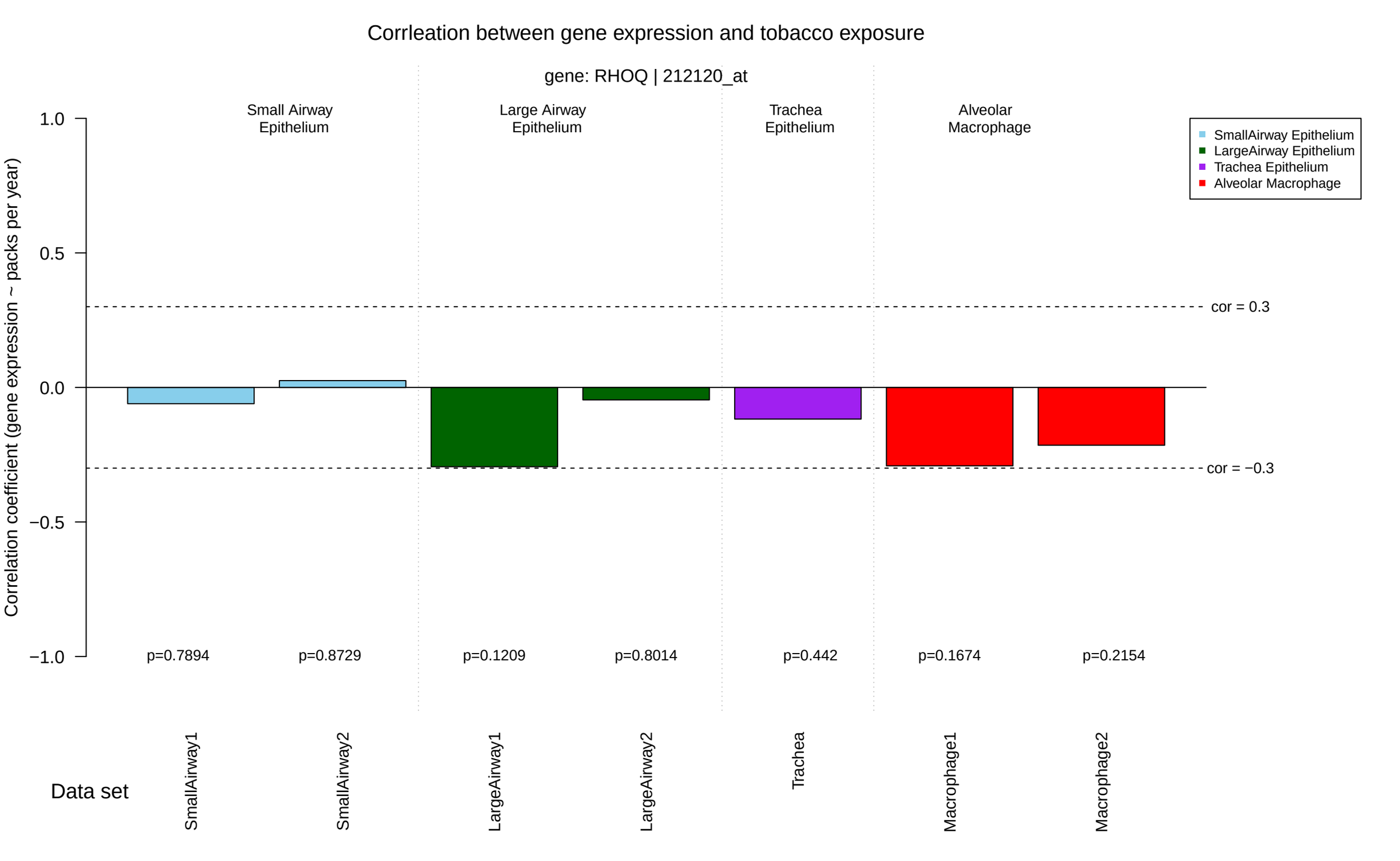

set: LargeAirway1 (p=0.0368)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7177)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0258)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0062)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0984)
RHOQ | 212120_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOQ | 212120_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7894
p=0.8729
p=0.1209
p=0.8014
p=0.442
p=0.1674
p=0.2154
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2